

Test Cases and Simulation Demonstration

A. Description of System Components

The smart lock system is composed of the following major components, each performing a specific function in the overall operation of the system:

- **RFID Reader (RC522):**
The RFID reader is used to perform contactless authentication. Each RFID card has a unique identification number (UID), which is read by the RFID reader and sent to the microcontroller. The UID is compared with the list of authorized UIDs stored in the system. If the UID matches, access is granted; otherwise, access is denied.
- **4×4 Keypad:**
The keypad is used as a secondary authentication method and for system configuration. Users can enter a password to unlock the door if RFID is unavailable. The keypad is also used by the owner to change the user password using a master password. This simulates real-world smart lock behavior where administrative access is protected.
- **Arduino Microcontroller (Simulated):**
The microcontroller acts as the central processing unit of the system. It receives inputs from the RFID reader and keypad, processes authentication logic, manages different operating states (normal mode and password configuration mode), and controls the output devices such as the servo motor, LEDs, and buzzer.
- **Servo Motor (Door Lock Mechanism):**
The servo motor simulates the physical door locking mechanism. When valid authentication is detected, the servo rotates to the unlock position. After a predefined delay, the servo automatically returns to the locked position, demonstrating the auto-lock feature.
- **LED Indicators (Green and Red):**
The green LED indicates successful authentication and access granted. The red LED indicates failed authentication and access denied. These LEDs provide immediate visual feedback to the user.

- **Buzzer:**

The buzzer provides audio feedback for failed authentication attempts, alerting the user and simulating an alarm or warning sound in case of unauthorized access.

B. Simulation Environment

The complete system is simulated using online simulation platforms.

- **Tinkercad** is used for initial logic verification and basic control testing.
- **Wokwi** is used for hardware-level simulation, including the RFID reader (RC522), keypad, servo motor, LEDs, and buzzer.

The simulation environment allows testing of all system features without physical hardware and provides real-time visualization of system behavior.

C. Test Cases Performed and Recorded in Video

Test Case 1: Authorized RFID Access (Blue Card)

- **Action:** Blue RFID card is tapped on the RFID reader.
- **Expected Result:**
 - UID matches authorized UID
 - Access granted
 - Green LED glows
 - Door unlocks and automatically locks after 5 seconds
- **Observed Result:**
 - Access granted
 - Green LED glows
 - Door unlocks and auto-locks after 5 seconds
- **Status:** Pass

Test Case 2: Authorized Keypad Access (Initial Password: 2580)

- **Action:** User enters 2580 using the keypad.
- **Expected Result:**
 - Access granted
 - Green LED glows
 - Door unlocks and automatically locks after 5 seconds
- **Observed Result:**
 - Access granted
 - Green LED glows
 - Door unlocks and auto-locks after 5 seconds
- **Status:** Pass

Test Case 3: Unauthorized Keypad Access (Wrong Password)

- **Action:** User enters an incorrect password.
- **Expected Result:**
 - Access denied
 - Red LED glows
 - Buzzer sounds
- **Observed Result:**
 - Access denied
 - Red LED glows
 - Buzzer sounds
- **Status:** Pass

Test Case 4: Unauthorized RFID Access (Wrong Green Card)

- **Action:** Unauthorized RFID card (green card) is tapped.
- **Expected Result:**
 - Access denied
 - Red LED glows
 - Buzzer sounds
- **Observed Result:**
 - Access denied
 - Red LED glows
 - Buzzer sounds
- **Status:** Pass

Test Case 5: Password Reset Using Master Password

- **Action:**
 - Master password is entered
 - New password is set
- **Expected Result:**
 - System enters configuration mode
 - User password is updated successfully
- **Observed Result:**
 - Password updated successfully
- **Status:** Pass

Test Case 6: Access Using New Password

- **Action:** User enters the new password.
- **Expected Result:**
 - Access granted
 - Green LED glows
 - Door unlocks and automatically locks after 5 seconds
- **Observed Result:**
 - Access granted
 - Green LED glows
 - Door unlocks and auto-locks after 5 seconds
- **Status:** Pass

Test Case 7: Access Using Old Password After Reset

- **Action:** User enters the old password (2580) after reset.
- **Expected Result:**
 - Access denied
 - Red LED glows
 - Buzzer sounds
- **Observed Result:**
 - Access denied
 - Red LED glows
 - Buzzer sounds
- **Status:** Pass

D. Summary of Simulation Results

The simulation results confirm that the smart lock system functions as intended. Both RFID-based and keypad-based authentication methods work correctly, unauthorized access attempts are properly rejected with audio-visual alerts, and the door consistently auto-locks after 5 seconds following any successful unlock. The password reset functionality using a master password operates reliably, ensuring secure configuration control. The recorded simulation video demonstrates all key behaviors of the system, validating the correctness of the design and implementation.